

Intellihealth – A Secured Decentralized Electronic Health Record System using Blockchain

Sangeeta Parshionikar

Department of Computer Engineering
Fr. Conceicao Rodrigues College of
Engineering
Mumbai, India
sangeeta@frcrce.ac.in or (0000-0003-
4940-8704)

Aarush Verulkar

Department of Computer Engineering
Fr. Conceicao Rodrigues College of
Engineering
Mumbai, India
info.aarushv@gmail.com

Ashay Katkar

Department of Computer Engineering
Fr. Conceicao Rodrigues College of
Engineering
Mumbai, India
katkarashay@gmail.com

Abstract— Intellihealth is an electronic health record system that operates online with a decentralized structure, setting it apart from conventional systems that store data locally. This decentralized approach relies on blockchain technology to ensure the security of patient data. Within this system, patients can easily access their medical information, including their history, prescriptions, and recommendations, but they are not granted editing privileges. Doctors, on the other hand, can add reports, access patient medical histories, and offer medical advice. Every transaction requires the consent of both parties involved. The system's owner, typically a hospital representative, has the authority to add patients and doctors to the database. Upon logging in, patients are assisted by a chatbot to navigate the system effectively. Additionally, a predictive analysis system featuring various machine learning models has been integrated. This allows patients to upload medical reports, such as X-rays and MRI scans, for the detection of diseases like pneumonia or brain tumors, providing them with a preliminary diagnosis. Intellihealth is a web application compatible with all major web browsers. It utilizes Angular with TypeScript for the front-end, IPFS for the database, Django with python for Back-end and blockchain software like Truffle and Ganache. Account sign-in on the decentralized platform is facilitated through a Metamask wallet and is linked to the user's Aadhar Number as the primary identifier.

Keywords— Blockchain, Electronic Health Record, Decentralized, Ethereum

I. INTRODUCTION

Over time, as patients seek medical care, an ever-increasing volume of physical documents and paper-based reports is generated. These records, comprising medical histories, test results, and treatment documentation, then become the patient's responsibility to maintain over the course of their treatment. Furthermore, should a patient decide to switch healthcare providers or facilities, they are burdened with the task of carrying these physical records to the new healthcare setting. Physical documents and reports are susceptible to loss or damage, thereby impeding the process of patient record accessibility and continuity of care. The risk of misplaced records or deterioration demands for a more secure and efficient means of managing patient data [1]. An electronic health record system, enhanced by blockchain technology, offers a promising solution to solve these challenges, ensuring the preservation and accessibility of patient records in a secure and convenient manner [2][3]. This system not only ensures the secure storage of patients' medical information but also facilitates the seamless transfer of these

records among registered healthcare professionals and institutions. By granting patients the autonomy to control access to their medical records, their privacy and confidentiality are preserved. An extensive repository is maintained that includes a thorough record of medical consultations and a range of diagnostic documents, including MRI scans and X-rays [4][5]. Crucially, this repository is easily accessible to patients, regardless of their location or the time of access. Furthermore, the immutability of blockchain technology ensures the safeguarding of unaltered patient histories, facilitating healthcare providers in the retrieval of medical records and their seamless addition with new data [6].

II. LITERATURE REVIEW

The research outlined in [6][7] proposed the implementation of a permissioned blockchain system for sharing and integrating Electronic Health Record (EHR) data. Within this setup, every hospital integrates a blockchain node connected to its individual EHR system, establishing an integrated blockchain network. To oversee data, a hybrid strategy was embraced, wherein solely management metadata found its place on the blockchain. The genuine EHR data underwent encryption and found storage off-chain within a cloud-based storage system that adhered to the Health Insurance Portability and Accountability Act (HIPAA). In another study [8], a Blockchain Handshake (BH) protocol was developed to facilitate connections between the database, the blockchain network, and the health record system, effectively serving as an interface between the components. [10] proposed approach integrates a dynamic reward system within the blockchain framework, where health providers play a crucial role in sustaining the integrity and functionality of the network. Furthermore, research case [11][12] introduces a consent-based dynamic architecture model for healthcare data sharing using blockchain technology, smart contracts, and ontologies for consent representation. Through the analysis of the dataset comprising data from wearable devices and smartphones of 29 patients, the authors assess the efficacy and scalability of the architecture across different contexts and scales of data sharing. The methodology outlined in reference [13][14] involves the creation of a prototype system designed to securely oversee patient records. Performance testing, which includes assessing the efficiency of chaincode logic, reveals that the system holds potential for adoption by medical practitioners [15].

III. METHODOLOGY

This section provides a formal description of the proposed frameworks. It outlines the software platform used to develop these frameworks and its advantages. The next section also describes the most prominent and crucial elements in implementing these frameworks - Ethereum and the Interplanetary File System (IPFS).

A. Protocol Layout

The figure 1 shows the layout of the system when a patient wants to view medical records using MetaMask or the electronic healthcare system's website. The user is automatically logged in using their private key which can be accessed from ethereum wallet. The wallet can be used for signing e-documents or for verification needs. Usage of private keys allows implementation of a role-based access system for stored records.

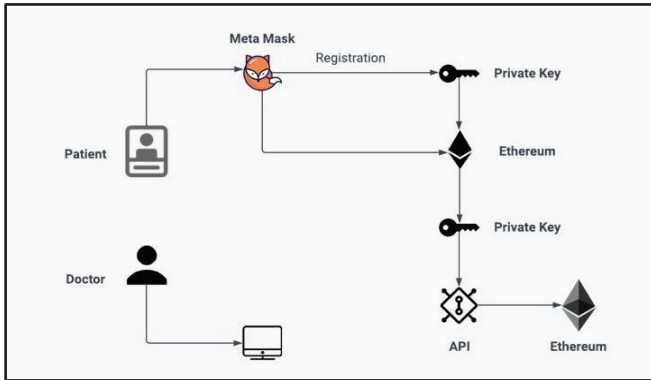


Fig. 1. System Layout.

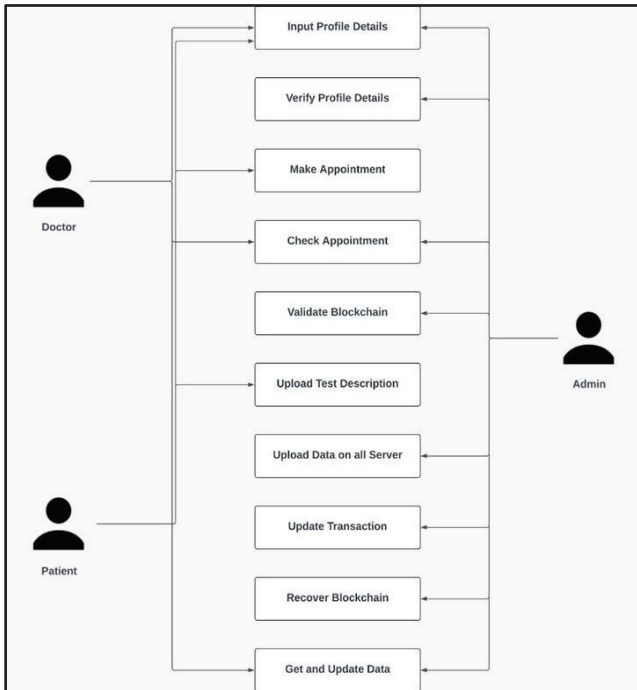


Fig. 2. Use Case Diagram.

B. Use of Case Diagram for Proposed System

The diagram 2 depicts our healthcare application with three user roles: administrator, doctor, and patient. Each role has specific access privileges. Patients have access to three functionalities out of ten, while doctors have access to three different functionalities. The administrator, with the highest level of access, can perform all ten actions, enabling comprehensive data monitoring. Notably, only doctors are authorised to modify patient record data stored on the blockchain, with such alterations being clearly marked as new transactions.

C. Admin Dashboard Process

The diagram 3 illustrates the workflow of the admin dashboard. Users accessing the system are identified through unique Ethereum wallet IDs. Upon successful ID verification, authorised accounts gain access to the admin dashboard. This dashboard empowers administrators with various functionalities, including adding and removing users, managing appointments, and monitoring booked appointments.

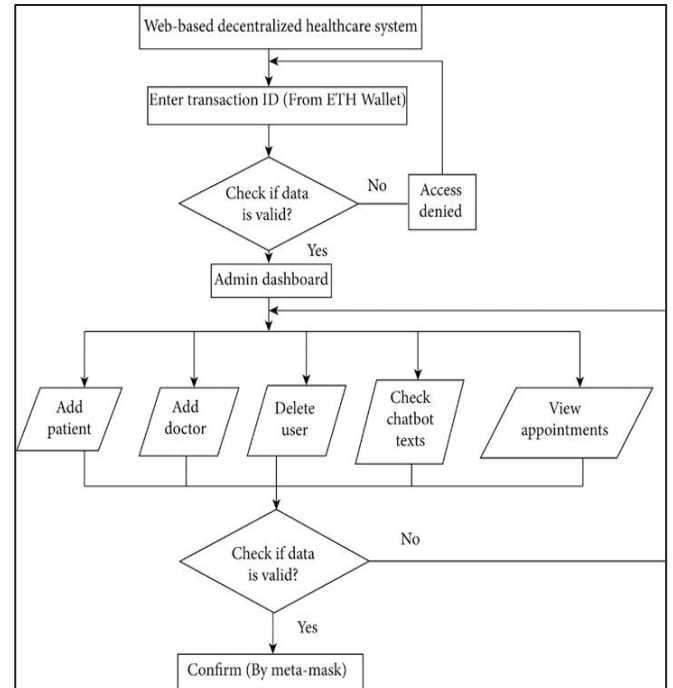


Fig. 3. Admin Workflow

D. Doctor Dashboard Process

The figure 4 showcases the doctor's dashboard layout. Doctors can be registered using a specific transaction ID and associated information. Access to this dashboard is restricted solely to authorised doctor accounts. Beyond appointment viewing capabilities, doctors have access to five additional functionalities. The dashboard allows doctors to review their personal information, including the option to update details like age, phone number, and photo. Additionally, doctors can access and examine patients' medical history to inform future treatment plans, with the ability to append further information as needed.

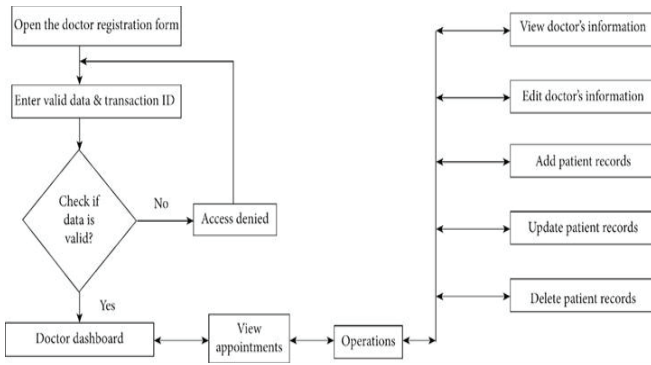


Fig. 4. Doctor workflow.

E. Patient Dashboard Process

The diagram outlines the patient dashboard workflow. Patients can register using a unique transaction ID and associated information. Access to this dashboard is strictly limited to verified patient accounts. Once registered, patients can schedule appointments and access their medical information. They can also view their personal details, with the ability to edit information such as name, age, phone number, address, and photo as needed.

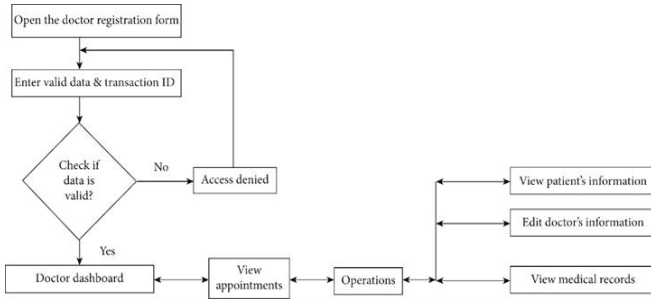


Fig. 5. Patient workflow.

IV. RESULTS

This research implements a decentralized system using Ethereum tools for constructing blockchains locally. Truffle, an integrated development environment, and Ganache, which generates accounts on a personal Ethereum blockchain, enable rapid prototyping and testing. The system's back-end logic utilises Solidity smart contracts programmed in conjunction with Node.js. To access the system as an end-user, one requires MetaMask, a virtual Ethereum wallet installed as a browser extension. Local Web3 provides a web interface for interacting with the contracts. Yarn manages dependencies and builds the app. By employing this toolkit for blockchain development, we abstract away low-level details while retaining fine-grained control over the full stack. Our system leverages Firebase, a decentralized storage solution integrated with the InterPlanetary File System (IPFS). This implementation provides end-to-end encryption for all sensitive health data prior to IPFS storage, guaranteeing that only authorized parties can access the information. This encryption process aligns with the stringent security requirements of both GDPR and HIPAA. This allows rapid implementation of a decentralized system with custom logic on a private test network, elucidating both the capabilities and limitations of blockchain architectures for peer-to-peer applications.

A. User Experience Flow

1) Homepage

As depicted in Figure 6, the system homepage serves as the gateway for users to access their respective portals after account creation. The homepage displays three portal options: system administrator, doctor, and patient. To log in, each user type (admin, doctor, patient) must register with unique credentials - the system denies access if incorrect or duplicate account information is provided. This secure access control enforces separation between the admin portal, doctor portal, and patient portal. After successfully authenticating, users can then utilise the full capabilities of their respective portal, which contains customised functions aligned with their user role.



Fig. 6. Application homepage

2) Appointment Panel

The appointment scheduling bar, shown in Figure 7, enables users to request and book appointments with doctors on the system. Patients can view doctor availability and select open time slots that work for their schedule. The chatbot feature, depicted in Figure 3, provides a convenient messaging interface for users to contact hospital staff. These incoming appointments are viewable by both doctors and administrators on their portals. This allows hospital staff to efficiently coordinate responses and bookings based on patient requests submitted through the system, rather than relying on external communication channels. The automated and transparent nature of the appointment bar and chatbot aims to simplify booking and administrative processes for both patients and healthcare providers.

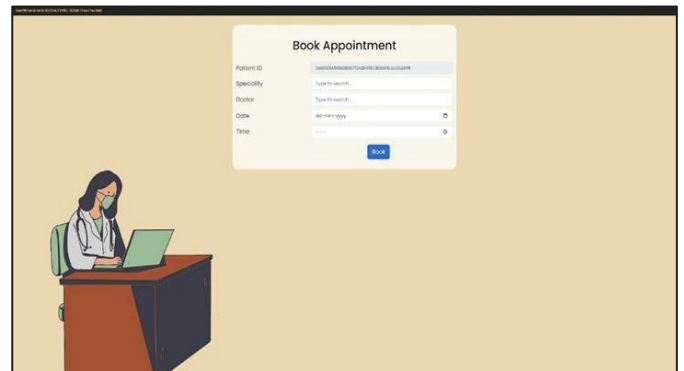


Fig. 7. Appointment page

3) Admin Panel

The administrator portal, shown in Figure 8, enables oversight of user accounts and system coordination. Core admin functions include registering new doctor and patient accounts, as well as revoking existing accounts if needed. The admin has a unique, unalterable account that cannot be accessed by other users. This privileged access reinforces security and authenticity.

The doctor registration process is depicted in Figure 8. As illustrated in Figure 8, the admin must enter the doctor's personal and professional details into the registration form. These details include name, date of birth, contact information, a unique doctor ID from Ganache, location, and medical specialty. Any incomplete or duplicate entries are denied, as shown by the access denied message. This rigorous registration protects against fraudulent accounts, ensuring only verified doctors can be added by the admin.

Fig. 8. Admin page.

4) Patient Registration Form

The process for the admin to register new patients is shown in Figure 9. Figure 9 displays the patient registration form requiring fields such as name, contact information, and a unique patient ID from Ganache. After fully entering the patient's data, the admin submits the form by clicking the "Add Patient" button, which initiates storing the information. Figure 12(b) shows the MetaMask confirmation that follows, implementing a smart contract to record the new patient's credentials on the blockchain ledger. MetaMask lists the Ethereum currency details used to immutably store the data. If any information is incomplete or duplicates an existing patient, the system denies access. Storing the data via MetaMask smart contract ensures security and authenticity of new patient accounts. The admin cannot register fake or redundant patient accounts, protecting against fraud. This meticulous patient registration process maintains integrity of the system by only allowing verified patients to be added by the administrator.

Fig. 9. Patient registration form.

5) Doctor's Panel

The doctor portal, shown in Figure 10, enables doctors to access and manage their account information and patient records. Key features include viewing their personal profile, adding new patients, and removing previous patients after completed consultations.

The process for doctors to view their information is illustrated in Figure 10. After identity verification via the blockchain, the doctor's profile details are displayed, including name, specialisation, qualifications and other attributes.

Fig. 10. Doctor Dashboard Page.

The doctor portal enables managing patient medical records and prescriptions as shown in Figure 11. Doctors can upload and update comprehensive records including personal details and medical history. Figure 11 shows the section for entering patient personal information like name, birthdate, contacts etc. Doctors can also add comments to establish rapport. The prescription form in Figure 12 lists diagnosed conditions, prescribed medications, recommended tests and side notes for the patient. Doctors can modify prescriptions by adding new drugs or tests based on evolving health needs. Unnecessary old records can be removed to reduce clutter. This doctor-exclusive access to add, modify and delete patient prescriptions supports efficient record management. The underlying blockchain technology guarantees data integrity and authenticity of all prescription iterations. The doctor portal balances usability with security for core EHR functions.

Fig. 11. Patient personal information.

Fig. 12. Addition of necessary drugs and tests.

6) Patient's Panel

The patient portal enables users to view their medical records as shown in Figure 6. Each patient has a unique ID corresponding to their health profile documented by doctors. The records display the date, time, and name of the consulting doctor for each visit. It categorizes diagnosed conditions such as heart disease, fever, or migraines. Based on the listed health issues, the doctor then prescribes medications. This provides complete transparency to patients regarding their evolving diagnoses and treatment plans over time. Patients have access to their full history of medical consultations and prescriptions, empowering them to take ownership of their health journey. Additionally, patients can print their records if needed for personal use or when consulting new providers. The ability to view their up-to-date medical history aims to help keep patients informed and engaged in managing their health. However, patients have read-only access to clinical records and prescriptions to prevent unauthorized modifications. This provides transparency into their complete health profile from diagnosis to medications.

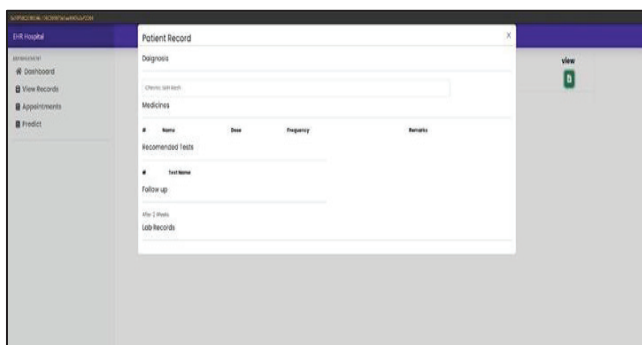


Fig. 13. Patient panel.

The patient portal allows users to view and update their personal information, as depicted in Figure 14. Patients first search for their unique ID, prompting identity confirmation via MetaMask. After blockchain verification, their demographic profile is displayed, including name, address, age, contacts, blood type, height, weight and other attributes. Patients can modify their details through the portal to keep them current. Any changes are securely recorded by saving the profile update using MetaMask's smart contract functionality. This allows patients to maintain accurate personal information while leveraging the blockchain to prevent unauthorized tampering. By enabling patients to control access to their information, the system aims to engage users in managing their health data while still ensuring security and authenticity via decentralization.



Fig. 14. Patient description

B. Discussions

After taking into consideration the increased demands from healthcare sector, analyzing the issues in existing solution, and our suggested blockchain-based healthcare management system in this paper, explaining new way of handling patient records and treatment planning, highlighting

drawbacks of existing solutions that hamper with personalized treatment, and giving an overview of how our proposed solution addresses these drawbacks.

It's important to note that our current tests were conducted on a testnet environment. While this provides valuable insights, we anticipate potentially improved performance on a mainnet due to more robust infrastructure in production environments, optimized network configurations and potential for increased network participants, leading to better distribution of workload.

Additionally, to ensure scalability and reliability, we can leverage cloud infrastructure for our performance testing. This setup will allow us to dynamically scale resources based on demand and measure performance across various instance types and configurations.

By leveraging blockchain characteristics to manage electronic health records, accessibility, coordination and security are few of the drawbacks that this system can resolve. Healthcare organizations are the primary target of malicious entities due to sensitive information they deal with. Usage of blockchain to manage electronic health records will protect the sensitive information from being tampered with or leaked for malicious use.

V. CONCLUSION

The amount of data generated within the healthcare sector is substantial. It is necessary to make enhancements to our existing data management practices without compromising the security and confidentiality of the data. Blockchain emerges as a viable solution to address the underlying fundamental issues.

The EHR system enables patients to authorize or revoke access to specific records for authorities through a single tap. The wallet functions as a secure bridge, ensuring convenient access to the blockchain and facilitating the hassle-free maintenance of user's secret keys. The system employs robust cryptographic encryption methods, ensuring unparalleled security and reliability. The achievement of the project's primary objectives, such as the secure exchange of authenticated medical reports using blockchain technology, is evident. Consequently, the successful implementation of the proposed methodology is anticipated to align with users' requirements of authentication, data exchange, and the security of medical reports through blockchain.

REFERENCES

- [1] Dubovitskaya A, Baig F, Xu Z, Shukla R, Zambani P, Swaminathan A, Jahangir M, Chowdhry K, Lachhani R, Idnani N, Schumacher M, Aberer K, Stoller S, Ryu S, Wang F, "ACTION-EHR: Patient-Centric Blockchain-Based Electronic Health Record Data Management for Cancer Care", J Med Internet Res 2020;22(8):e13598 URL: <https://www.jmir.org/2020/8/e13598> DOI: 10.2196/13598
- [2] Farjana Khanam Nishi, Mahizebin Shams-E-Mofiz, Mohammad Monirujjaman Khan, Abdulmajeed Alsufyani, Sami Bourouis, Punit Gupta, Dinesh Kumar Saini, "Electronic Healthcare Data Record Security Using Blockchain and Smart Contract", Journal of Sensors, vol. 2022, Article ID 7299185, 22 pages, 2022. <https://doi.org/10.1155/2022/7299185>
- [3] G. Yang and C. Li, "A Design of Blockchain-Based Architecture for the Security of Electronic Health Record (EHR) Systems," 2018 IEEE International Conference on Cloud Computing Technology and Science (CloudCom), Nicosia, Cyprus, 2018, pp. 261-265, doi: 10.1109/CloudCom2018.2018.00058.

- [4] V. Jaiman and V. Urovi, "A Consent Model for Blockchain-Based Health Data Sharing Platforms," in *IEEE Access*, vol. 8, pp. 143734-143745, 2020, doi: 10.1109/ACCESS.2020.3014565.
- [5] Dubovitskaya A, Xu Z, Ryu S, Schumacher M, Wang F. Secure and trustable electronic medical records sharing using Blockchain. *AMIA Annu Symp Proc* 2018;2017:650-9.
- [6] R. Neisse, G. Steri, and I. Nai-Fovino, "A blockchain-based approach for data accountability and provenance tracking," in *Proc. 12th Int. Conf. Availability, Rel. Secur.*, Aug. 2017, pp. 1–10, doi: 10.1145/3098954.3098958.
- [7] M. Vukolić, "The quest for scalable blockchain fabric: Proof-of-work vs. bft replication," in *Open Problems in Network Security*, J. Camenisch and D. Kesdoğan, Eds. Cham, Switzerland: Springer, 2016, pp. 112–125.
- [8] Magyar G., "A new disruptive technology in health data management. Blockchain Solving the privacy and research availability tradeoff for EHR data", *IEEE 30th Neumann Colloquium (NC)*; 2017; Budapest. [doi: 10.1109/nc.2017.8263269]
- [9] Pournaghi SM, Bayat M, Farjami Y., "MedSBA: a novel and secure scheme to share medical data based on blockchain technology and attribute-based encryption", *J Ambient Intell Human Comput* 2020 Jan 21:1-18. [doi: 10.1007/s12652-020-01710-y]
- [10] Paranjape K, Parker M, Houlding D, Car J., "Implementation Considerations for Blockchain in Healthcare Institutions", *BHTY 2019 Jul 04*;2:1-15. [doi: 10.30953/bhty.v2.114]
- [11] Azaria A, Ekblaw A, Vieira T, and Lippman A., "Medrec: Using blockchain for medical data access and permission management", In *2016 2nd International Conference on Open and Big Data (OBD)*, pages 25–30, Aug 2016
- [12] Yue X, Wang H, Jin D, Li M, and Jiang W. Healthcare data gateways: Found healthcare intelligence on blockchain with novel privacy risk control. *Journal of Medical Systems*, 40(10):218, 2016.
- [13] Beninger P and Ibara MA. Pharmacovigilance and biomedical informatics: a model for future development. *Clinical Therapeutics*, 2016.
- [14] X. Liang, J. Zhao, S. Shetty, J. Liu and D. Li, "Integrating blockchain for data sharing and collaboration in mobile healthcare applications," *2017 IEEE 28th Annual International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC)*, Montreal, QC, 2017, pp. 1-5
- [15] T. Kuo, H. Kim and L. Ohno-Machado, "Blockchain distributed ledger technologies for biomedical and health care applications", *Journal of the American Medical Informatics Association*, 2017, Vol. 24, No. 6.
- [16] F. Tschorsch and B. Scheuermann, "Bitcoin and Beyond: A Technical Survey on Decentralised Digital Currencies", *IEEE Communications Surveys Tutorials*, 2016, volume 18,3, 2048- 2123
- [17] N. Havelange, M. Dumontier, B. Wouters, J. Linde, D. Townend, A. Riedl, and V. Urovi, "LUCe: A blockchain solution for monitoring data license accountability and Compliance," 2019, arXiv:1908.02287. [Online]. Available: <https://arxiv.org/abs/1908.02287>